

Data Management Framework to the Operational and Execution Reporting Process Project

Group 7

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1. Introduction

1.1 Purpose:

This document outlines the data management framework for operational and executive reporting by defining governance principles, policies, and procedures.

1.2 overview:

This framework integrates elements from two subdomains of the DAMA wheel: metadata management and reference & master data management, ensuring data consistency, accuracy, and traceability.

1.3 Business Assumptions:

- The organization functions in a data-driven environment that demands real-time, structured, and accurate reporting for effective decision-making.
- Key stakeholders include executives, managers, analysts, and external regulators.
- Reports serve various purposes: operational reports track daily performance, while executive reports offer strategic insights.
- To facilitate effective reporting, the organization utilizes modern data architectures, cloud-based storage, and metadata repositories.
- Reports should include automated data validation checks to guarantee completeness and reliability.
- Implementation of role-based access control is essential to ensure that only authorized personnel have access to sensitive report data.

2. Selected DAMA Sub-Domains

2.1 Metadata Management

Metadata offers crucial context about data, ensuring consistency, traceability, and standardization across reports. It improves data lineage and assists stakeholders in accurately interpreting reports.

2.2 Reference & Master Data Management

Master and reference data standardize essential business entities like customers, products, and locations, ensuring accuracy and consistency across all reports.

3. Principles, Policies and Procedures

3.1 Metadata Management Principles

3.1.1 Metadata Documentation

Principle: Reports must incorporate documented metadata to ensure proper traceability and facilitate accurate interpretation.

Policies:

1. **Metadata Entry Requirements:** Each report field must include a metadata entry that defines its source, calculation method, and business purpose.
 - a. **Procedure:**
 - i. Report developers must use a standardized template to input metadata.
 - ii. Metadata must include details such as name, description, and data owner.
 - iii. Submit the metadata entry to the centralized repository
 - iv. Metadata steward reviews for compliance.
2. **Metadata Review Process:** All metadata updates must be reviewed and approved by the Data Governance team prior to implementation.
 - a. **Procedure:**
 - i. Automated scripts validate metadata changes.
 - ii. Data Governance team reviews flagged discrepancies.
 - iii. Approved metadata updates are logged in a revision history.

3.1.2 Metadata Accessibility

Principle: It is essential to store metadata in a readily accessible repository to empower stakeholders to verify data lineage and transformation processes effectively.

Policies:

1. **Centralized Metadata Repository:** A centralized repository for metadata will be maintained, granting read access to all users of the reports.
 - a. **Procedures:**

- i. Establish a centralized metadata repository with secure access controls.
- ii. Grant read-only access to all authorized users based on role-based permissions.
- iii. Ensure metadata repository is regularly synchronized with reporting tools.
- iv. Schedule periodic audits to verify metadata accessibility compliance.

2. Metadata Access Controls: Access to metadata should be regulated to prevent any unauthorized modifications.

a. Procedures:

- i. Implement role-based permissions for metadata repository access.
- ii. Restrict write permissions to designated metadata stewards and administrators.
- iii. Enable logging of metadata access and modification activities.
- iv. Conduct quarterly security reviews to ensure compliance with data governance policies.

3.2 Reference & Master Data Management Principles

3.2.1 Standardized Reference Data

Principle: Reports should utilize standardized reference data that aligns with the master dataset to ensure consistency across reports.

Policies:

1. Master Table Integration: All product and customer references in reports must correspond to the master tables.

a. Procedures:

- i. Identify and define the required master data fields (e.g., product categories, customer segments).
- ii. Implement foreign key constraints in the database to enforce consistency.
- iii. Regularly validate reference data against the master data repository.
- iv. Automate synchronization of master tables across reporting systems.

2. Reference Data Validation: All reference data used in reports must be validated before publication.

a. Procedures:

- i. Establish validation rules for each reference data type (e.g., country codes, product categories).
- ii. Implement automated validation scripts to detect inconsistencies or missing values.
- iii. Conduct periodic audits to compare reference data with external authoritative sources.
- iv. Flag discrepancies and notify data stewards for corrective action before reports are published.

3.2.2 Secure Access to Master Data

Principle: Role-based security measures should be implemented to safeguard sensitive master data from unauthorized changes.

Policies:

1. **Role-Based Access Control for Master Data:** Only authorized users should have permission to modify master data records.

a. Procedures:

- i. Define access levels for different user roles (e.g., read-only, editor, administrator).
- ii. Assign role-based access to master data systems through a centralized access control system.
- iii. Restrict write access to designated data stewards and master data managers.
- iv. Conduct quarterly access reviews to ensure compliance with security policies.

2. **Master Data Access Logging & Monitoring:** All access and modifications to master data must be logged and monitored for security compliance.

a. Procedures:

- i. Enable logging for all master data access and modifications in the system.
- ii. Store access logs in a secure, centralized repository for audit purposes.
- iii. Implement real-time monitoring to detect unauthorized access attempts.
- iv. Schedule regular audits to review access logs and identify potential security risks.

4. Conclusion

This framework ensures that reporting is well-controlled, transparent, efficient, and focused on quality by establishing consistent metadata documentation and effective master data governance. This approach reduces discrepancies and improves data accuracy. Regular audits, validations, and metadata tracking promote long-term data integrity, ensuring that reporting processes remain reliable and trustworthy.